Auditing the Architecture of Global Health Estimates: Bias, Artificial Smoothness, and a Reliability Lens

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A forensic data science analysis of reliability, resilience, and uncertainty in WHO mortality metrics

1 Abstract

Background. WHO Global Health Estimates (GHE) are widely used for policy and monitoring. Where civil registration (CRVS) data exist, comparing reported and estimated mortality can clarify how interpretable and reliable estimates are, but few frameworks do this in a structured, reproducible way.

Objectives. To define an audit universe for CRVS–GHE comparison, quantify divergence and structural smoothness, classify volatility–divergence regimes, and produce a tiered reliability lens that third parties can use without running the pipeline.

Methods. We built a single panel of reported (WHO Mortality Database) and estimated (GHE) all-cause deaths (2000–2021, ICD-10). We computed bias and divergence; an Artificial Smoothness Index (ASI = roughness of reported / roughness of estimated on log-deaths) to detect when estimated series are structurally smoother than reported; and a volatility–divergence quadrant (median split). A reliability tier (A–D) was assigned per country using transparent rules (quadrant, ASI thresholds 1.2 and 1.5, high-divergence flag). Time-series metrics required $\geq$5 consecutive years (T). Time-series eligibility is parameterised ($T \in \{3, 5, 8\}$); results presented here use the default $T = 5$. Framework is descriptive and non-inferential.

Results. In the pilot, seven countries had eligible time series. Tier A (robust): 3; Tier C (use with caution): 1; Tier D (high concern): 3. Four countries had ASI > 1. Country cards and a dataset card were produced.

Conclusions. The framework is reproducible. It supports reliability-aware interpretation of estimates where CRVS and GHE coexist, without judging model validity. ASI and the tier system are the main operational contributions.

Keywords: global health estimates, mortality, CRVS, WHO, artificial smoothness, reliability, diagnostic framework, data quality, governance.

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2 Executive Summary

Purpose. This report provides a diagnostic framework for comparing reported mortality (CRVS, via the WHO Mortality Database) with WHO Global Health Estimates (GHE) where both

exist. The aim is to support *reliability-aware* use of estimates: making divergence, structural smoothness, and volatility explicit so that users and producers can apply appropriate caveats and align on what "use with caution" means. We do not judge whether GHE models are right or wrong; we supply a *lens* for interpretation, not an exposure or critique.

What we do. We define an audit universe (country-years with ICD-10 mortality and population data, 2000–2021), join reported and estimated all-cause deaths, and compute four linked outputs: (i) **bias and divergence** (level, trend, stability of reported vs estimated); (ii) an **Artificial Smoothness Index (ASI)**, the ratio of roughness of the reported series to roughness of the estimated series on log-deaths, so that ASI > 1 indicates the estimated series is structurally smoother than reported; (iii) a **volatility–divergence classification** (quadrants Q1–Q4 by median split: high/low volatility of reported series $\times$ high/low divergence); and (iv) a **reliability tier (A/B/C/D)** per country, with explicit rules and a "use with caution" flag. All logic and thresholds are documented; country cards and a dataset card allow third parties to use the lens without running the pipeline.

What we do not do. We do not assess model validity, propose new WHO governance arrangements, or attribute observed patterns to specific causes (e.g. modelling vs data quality). The framework is descriptive and non-inferential.

Pilot results. Seven countries had sufficient consecutive years ($T \geq 5$) for time-series diagnostics. Three are Tier A (robust: low volatility, low divergence, ASI ≤ 1.2); one is Tier C (use with caution); three are Tier D (high concern: high volatility and divergence, ASI > 1.5). Four of seven countries show ASI > 1. Results are illustrative of the framework; scaling to more countries is planned.

Audience and use. The report is for policymakers, analysts, and researchers who use or produce global health estimates and need a structured way to assess reliability when reported and estimated data coexist. The tier system and country cards are designed for citation and operational use in reporting and dialogue.

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3 Why Reliability Matters

Global health estimates fill gaps where civil registration is incomplete or missing. They are essential for monitoring progress, allocating resources, and comparing populations. Yet in countries that *do* report mortality, reported (CRVS) and estimated (GHE) series can differ in level, trend, and volatility. These differences are not necessarily errors; they may reflect different definitions, modelling assumptions, or adjustments, but they affect how both series should be interpreted and communicated.

A **reliability lens** does two things. First, it makes divergence and related diagnostics explicit, so that users can apply appropriate caveats. Second, it supports **governance**: transparent, reproducible diagnostics help data producers and consumers align on what "use with caution" means and where extra scrutiny is warranted. This report does not prescribe policy; it provides a methodological and operational framework that can be cited and extended.

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4 Data Sources: CRVS vs GHE

Reported mortality (CRVS). We use the WHO Mortality Database: cause-of-death counts and population by country, year, sex, and cause (ICD-10, List 101). Data availability is taken

from the official WHO availability file (e.g. February 2025). We do not treat missing country-years as zero deaths; missingness is explicit. National-level, all-cause deaths are obtained from the standard all-cause code (List 101); population is taken from the WHO population file for the same country-year where available.

Estimated mortality (GHE). We use WHO Global Health Estimates: country-level all-cause death counts (and, where applicable, cause-specific) for 2000–2021. GHE are modelled estimates; methods are documented in WHO technical reports (e.g. GHE 2021 cause-of-death methods). We use the same time window and geography as the CRVS layer so that joins are aligned.

Join. Reported and estimated series are joined on country (ISO3), year, sex, and cause group. In this version we restrict to **all-cause** mortality; cause-specific and DALY extensions are noted as future work. The join table and schema are documented in the project repository (METHODOLOGY/03_join_schema_reported_vs_estimated.md).

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5 Audit Universe (Stage 0)

The audit universe is defined so that exclusions are traceable and consistent across all downstream stages.

Geography and time. We include all countries and territories for which the WHO Mortality Database contained at least one year of cause-of-death data coded under ICD-10 (List 101) between 2000 and 2021. The time window aligns with the latest GHE country-level series and spans pre-SDG baseline through the COVID-19 period.

Inclusion. A country-year is included in the main audit if (i) ICD-10 mortality data are present for that country and year, and (ii) reference population data are available for the same country-year, so that completeness and rate checks can be performed. Country-years with mortality data but without population data are retained in the availability matrix but excluded from rate-based and divergence calculations that require denominators.

Exclusion. A country-year is excluded if it is not listed in the availability file as having ICD-10 mortality data. We never treat "no row" as zero deaths.

High-quality subset. For sensitivity, we also define a subset with at least three consecutive years of ICD-10 data within 2000–2021. Time-series metrics (bias stability, ASI, volatility) are computed only on countries with at least **T** consecutive years in the matched panel. Time-series eligibility is parameterised ($T \in \{3, 5, 8\}$); results presented here use the default $T = 5$. Full specification: METHODOLOGY/02_Stage0_inclusion_exclusion_rules.md.

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6 Forensic Framework

The framework has three measurement pillars (bias, artificial smoothness, volatility) and one synthesis (reliability lens). All are descriptive and non-inferential. Figure 1 shows the pipeline from audit universe through to the reliability lens.

6.1 Unit of analysis and eligibility

We build a single analysis panel with one row per (country_code, year, sex, cause_group) where both reported and estimated all-cause deaths exist. Time-series features are computed only for countries with at least T consecutive years in this matched panel (default $T = 5$). Cause scope in this version is **all-cause only**; cause-specific and DALY are not implemented in v1.

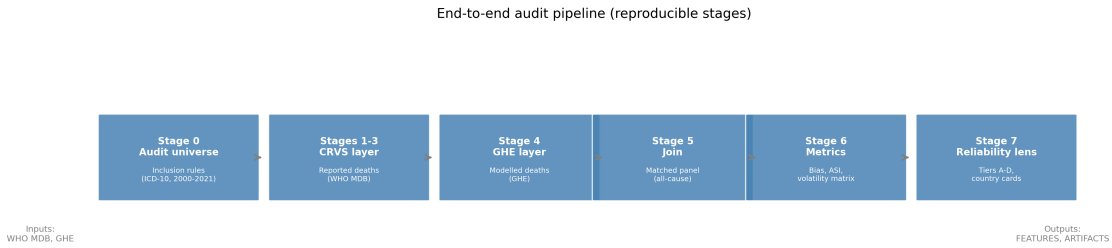

Figure 1: Pipeline: audit universe → CRVS → GHE → join → panel → bias / ASI / volatility → reliability lens (Stages 2–7; Stages 0–1 define design and inputs).

6.2 Bias and divergence (Stage 6.1)

We compute pointwise and country-level divergence between estimated and reported deaths: absolute and relative gaps (estimated − reported), ratio and log-ratio, mean absolute percent difference (MAPD), and the share of years in which the ratio exceeded a chosen threshold. "Bias stability" is the standard deviation of log-ratio across years: high values indicate that the direction of divergence is not consistent over time. Countries and years with the largest divergence are identifiable from the feature tables and QA reports (FEATURES/bias_signals.parquet, REPORTS/04A_bias_signals.md). We do not assign causal interpretation to "bias"; we use the term to mean observed systematic difference in level or trend.

6.3 Artificial Smoothness Index (Stage 6.2)

Modelled series are often smoother than reported series because of smoothing assumptions or aggregation. We quantify this with an **Artificial Smoothness Index (ASI)**. For each eligible country we work with the annual series of **log(deaths** $+ \varepsilon$**)** (deaths, not crude rates). We define **roughness** R as the median of the absolute second discrete differences of this series. We compute R for the reported series (R_{rep}) and for the estimated series (R_{est}), and define:

$$\text{ASI} = R_{\text{rep}}/(R_{\text{est}} + \varepsilon).$$

- ASI > 1: the estimated series is smoother than the reported series (potential artificial smoothness).
- ASI ≈ 1: similar smoothness.
- ASI < 1: estimated series is rougher than reported (rare).

Results are stored in FEATURES/artificial_smoothness_index.parquet and summarised in REPORTS/04B_artificial_smoothness.md. This metric is sensitive to series length and to outliers; we report it with caveats for small n.

6.4 Volatility matrix (Stage 6.3)

We classify countries into volatility–divergence regimes to support the reliability lens. For each eligible country we compute:

- **x-axis (volatility of reported series):** volatility_reported = standard deviation of the first differences of log(deaths).
- **y-axis (divergence magnitude):** divergence_magnitude = median(|log_ratio|) over years.

Quadrants are assigned by **median split** (descriptive partition; not inferential):

- **Q1:** High volatility, high divergence (red flag); reported series is volatile and diverges strongly from estimates.

- **Q2:** High volatility, low divergence: model tracks chaotic reported series reasonably well.
- **Q3:** Low volatility, high divergence: apparent smoothing or systematic bias.
- **Q4:** Low volatility, low divergence: stable and aligned.

We state caveats for small-n and for median-split behaviour. Results: FEATURES/volatility_matrix_labels.par REPORTS/04C_volatility_paradox.md. Figure 2 illustrates the volatility–divergence space for the pilot countries.

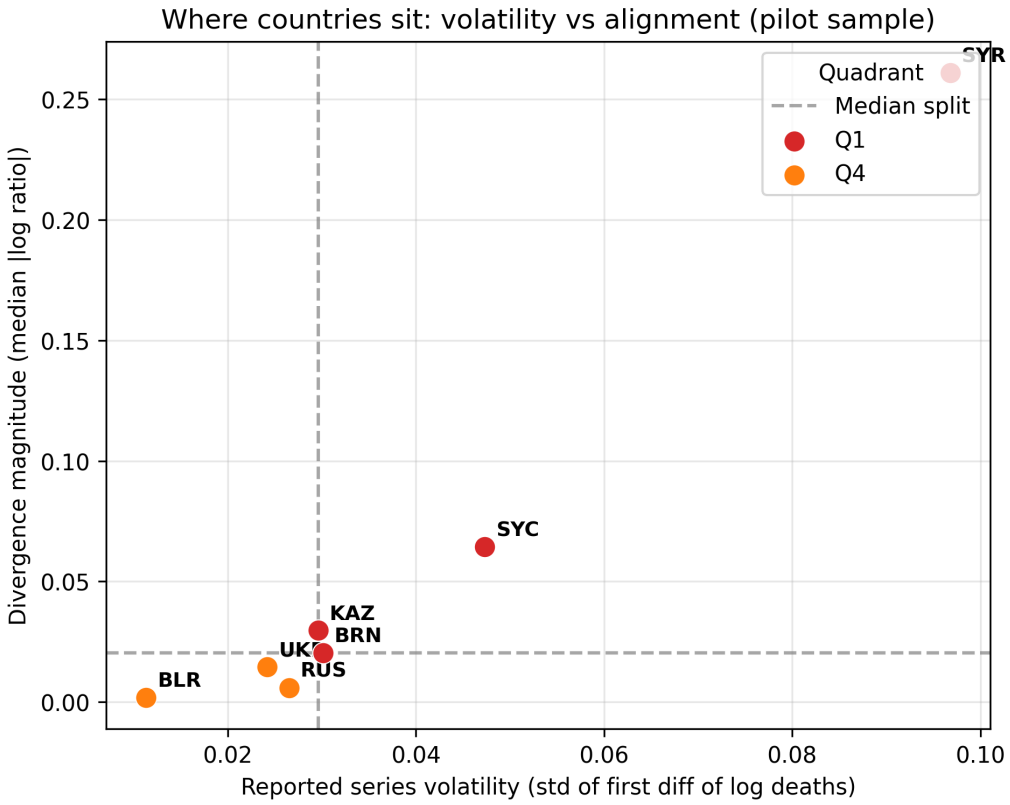

Figure 2: Volatility matrix: volatility of reported series (x) vs divergence magnitude (y); quadrants Q1–Q4 by median split. Country codes labeled.

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7 Reliability Lens (Stage 7)

The reliability lens converts the diagnostics above into a single, interpretable tier per country and a "use with caution" flag. A third party can use the lens without running the pipeline.

Inputs. One row per eligible country from: ASI table, volatility/quadrant table, and country-level bias aggregates.

Tier rules (v1). Tiers are assigned by evaluating conditions in order **D** → **C** → **B** → **A**; the first matching tier is applied.

- **Tier D (high concern):** Quadrant Q1 (high volatility, high divergence) **and** ASI > 1.5. Red-flag volatility and strong artificial smoothness; use with caution.
- **Tier C (use with caution):** Any of: Q1; or ASI > 1.5; or Q3 (low volatility, high divergence). Notable smoothing and/or systematic divergence.
- **Tier B (moderate):** Q2 or Q4 with moderate ASI (e.g. ASI in (1.2, 1.5]) or high divergence by median split. Context-dependent.

- **Tier A (robust):** Q4 **and** ASI $\leq$ 1.2 **and** divergence below median. Low concern; estimates align with reported series; plausible volatility.

Thresholds. ASI "high" > 1.5; ASI "moderate" > 1.2; "high divergence" = above median divergence_magnitude among eligible countries. If ASI is missing (e.g. too few years), tier is based on quadrant and divergence only, with "asi_missing" noted in reasons.

Outputs. ARTIFACTS/reliability_lens.csv (country_code, tier, quadrant, asi, divergence_magnitude, main_reasons, use_with_caution, n_years); ARTIFACTS/country_cards/<ISO3>.md (one-pager per country); ARTIFACTS/dataset_card.md. Full tier logic: METHODOLOGY/05_Stage7_reliability_tier_l

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8 Results

Audit universe and panel. The analysis panel (FEATURES/panel_allcause_matched.parquet) contains one row per (country_code, year, sex) with both reported and estimated all-cause deaths. In the current run, **84** country-year observations have both series; **9** countries have at least one such year. Time-series eligibility ($\geq$ 5 consecutive years with both) yields **62** country-years and **7** countries. All downstream diagnostics (bias, ASI, volatility, reliability tier) are computed on these 7 countries.

Bias and divergence. Country-level divergence (median log-ratio, MAPD, share of years with ratio > 1.1) is computed and reported in REPORTS/04A_bias_signals.md. Under-estimation (GHE < reported) and over-estimation (GHE > reported) are both observed; magnitude and stability vary by country. Highest divergence is seen in countries with volatile reporting or small n; we do not infer causation.

Artificial Smoothness Index. Among the 7 countries with ASI, **4** have ASI > 1 (estimated smoother than reported). Median ASI is 1.13; the distribution is right-skewed, with a few countries showing very high ASI (e.g. 5–15). Details: REPORTS/04B_artificial_smoothness.md. Figure 3 shows reported vs estimated all-cause deaths for one Tier A country (BLR) and one Tier D country (SYR), illustrating the contrast in smoothness.

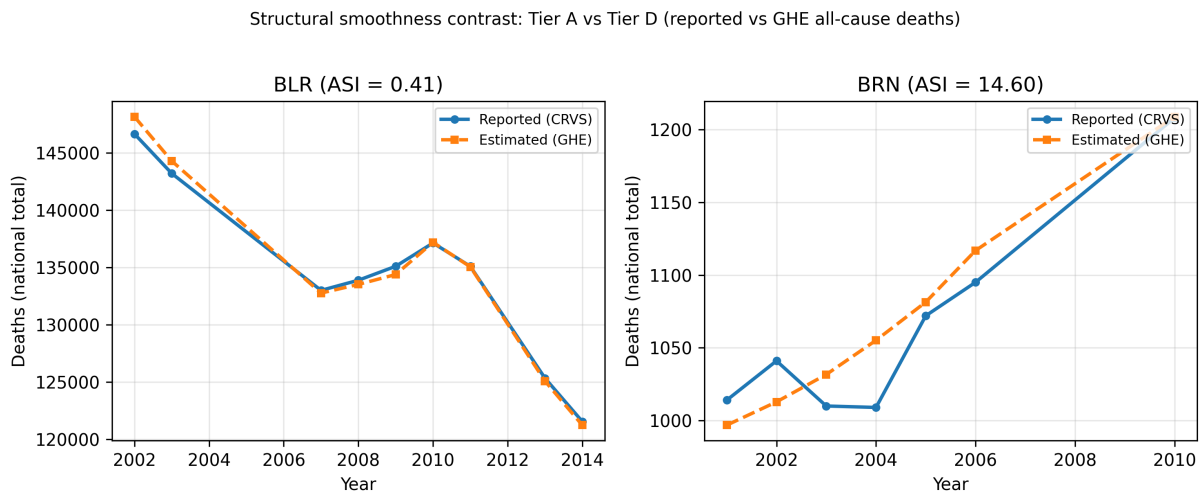

Figure 3: Reported vs estimated all-cause deaths: one Tier A and one Tier D country (ASI annotated).

Volatility matrix. Quadrant counts in this run: **Q1** (high volatility, high divergence) 4 countries; **Q4** (low volatility, low divergence) 3 countries; Q2 and Q3 empty in this small sample.

REPORTS/04C_volatility_paradox.md gives per-country values and quadrant definitions.

Reliability lens. Tier counts: **A = 3**, **B = 0**, **C = 1**, **D = 3**. Four countries are flagged "use with caution" (True); three are not. Example: one Tier A country (e.g. BLR) shows Q4, ASI $\approx$ 0.41, low divergence; one Tier D country (e.g. SYR) shows Q1, ASI $\approx$ 5.1, high divergence and is flagged. Full lens: ARTIFACTS/reliability_lens.csv; one-pagers: ARTIFACTS/country_cards/; QA: REPORTS/05_reliability_lens_QA.md. Figure 4 summarises the pilot tier counts.

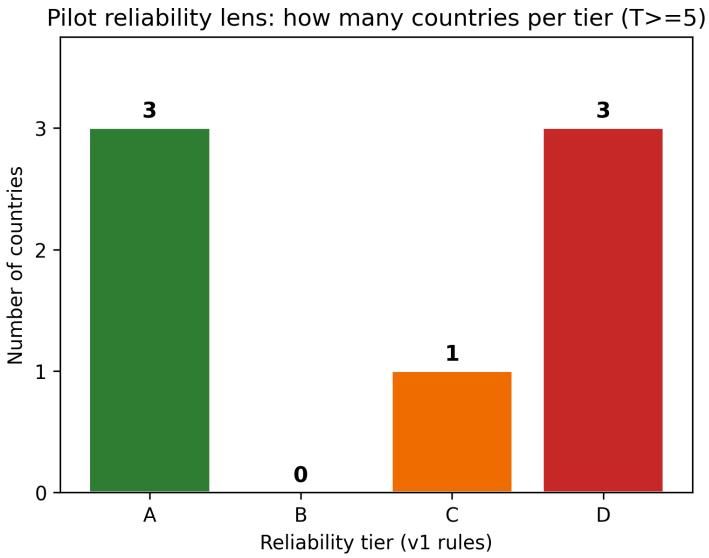

Figure 4: Pilot reliability lens (7 countries, T $\geq$ 5). Tier definitions in Methods.

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9 Interpretation

The framework is **diagnostic**, not evaluative. Observed divergence may reflect data quality, modelling choices, or definitional differences; we do not attribute cause. High ASI indicates that the estimated series is smoother than the reported series in a mathematical sense; it does not by itself imply that the model is "wrong." Similarly, Tier D flags conditions that warrant caution when interpreting estimates, not a verdict on the underlying methodology.

Governance use. Producers and users of estimates can use the reliability lens to (i) identify countries where caveats should be stated explicitly, (ii) prioritise validation or dialogue, and (iii) communicate uncertainty in a consistent way. The tier system and country cards are designed to be stable and reproducible so that they can be cited and updated as data and methods evolve.

Small-n caveat. With seven countries in the current pilot, quadrant and tier counts are illustrative. Rankings and tier assignments can change when more countries enter the audit universe or when T (consecutive years) is varied. Time-series eligibility is parameterised (T $\in \{3, 5, 8\}$); results presented here use the default T = 5. Extension to more countries and to cause-specific and DALY metrics is planned as future work.

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10 Limitations

- **Scope.** All-cause mortality only in v1; cause-specific and DALY are not included. The audit universe is limited to countries and years present in the WHO Mortality Database

with the required population and GHE coverage.

- **Eligibility.** Time-series metrics (ASI, volatility, tier) require at least T consecutive years (default 5); countries with sparse reporting are excluded from the lens.
- **Descriptive only.** No inferential statistics or causal claims. Median splits and thresholds (e.g. ASI 1.2, 1.5) are chosen for transparency and reproducibility, not from a formal optimisation.
- **Data versions.** Results depend on the specific releases of the WHO Mortality Database and GHE used; reproducibility requires documenting and, where possible, versioning these inputs.
- **No policy prescription.** The framework does not recommend specific governance or reporting changes; it provides a tool for reliability-aware interpretation.

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11 Governance Implications

Transparent, reproducible diagnostics support **evidence-informed governance** of global health estimates. By making divergence, smoothness, and volatility explicit, the framework allows:

- **Users** to apply appropriate caveats when citing or using estimates in high-divergence or high–ASI settings.
- **Producers** to identify where estimates and reported data diverge and to communicate limitations consistently.
- **Stakeholders** to align on what "use with caution" means operationally, without conflating diagnostic signals with normative judgement.

This report does not propose new WHO or intergovernmental processes; it contributes a methodological and operational lens that can be adopted, adapted, or cited by any party working with reported and estimated mortality data.

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12 Reproducibility and Data Availability

Code and logic. The pipeline is implemented in Python scripts for Stages 2–7 (with Stages 0–1 defining design and inputs) and documented in the project repository. Key documents: MASTER_ROADMAP_STAGE_WISE_SUMMARY.md, METHODOLOGY/ (Stage 0 rules, join schema, Stage 6 methods snippet, Stage 7 tier logic), configs/constants.py (e.g. default $T = 5$; $T \in \{3, 5, 8\}$ parameterises time-series eligibility).

Data. Input data are from the WHO Mortality Database (CRVS, availability file) and WHO Global Health Estimates (country-level deaths). Use of these sources is subject to their respective terms and citations; the repository documents file names and versions used (e.g. AVAILABILITY_list_ctry_years_feb2025.xlsx, GHE country-level series 2000–2021).

Outputs. FEATURES/ (panel, bias_signals, artificial_smoothness_index, volatility_matrix_labels), ARTIFACTS/ (reliability_lens.csv, country_cards/, dataset_card.md), REPORTS/ (04A, 04B, 04C, 04_STAGE6_PANEL_QA, 05_reliability_lens_QA). Figures: FIGURES/fig01_pipeline.png, fig02_country_comparison.png, fig03_volatility_matrix.png, fig04_tiers.png. Running the pipeline from Stage 2 through Stage 7 reproduces these outputs given the same inputs and constants.

How to run. See README and scripts/run_all.sh. A requirements file (requirements.txt) lists dependencies (e.g. pandas, pyarrow, openpyxl, pyyaml). Optional: use a virtual environment (e.g. `python3 -m venv .venv`; `.venv/bin/pip install -r requirements.txt`).

This work is an independent research effort and does not represent the views, decisions, or policies of the World Health Organization (WHO) or any affiliated institution.

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13 How to Cite

Recommended citation (APA)

Rawat, A. (2026). Auditing the Architecture of Global Health Estimates: Bias, Artificial Smoothness, and a Reliability Lens (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.19545271

BibTeX

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Suggested sentence for methods

We applied a forensic reliability framework (audit universe, bias and divergence, Artificial Smoothness Index, volatility matrix, reliability tier) as described in Rawat (2026) to compare reported and estimated all-cause mortality.